

Project #2

MyChat



MyChat

Administrative

Group size : 2

Repository name : mychat_{your-epita-email}

Example: mychat_johndoe@epita.com

Repository visibility : private

Constraints / recommendations

Libraries allowed:

- Material / Jetpack Compose
- [Fuel](#)

Android Studio highly recommended.

Environment

Android version : 16.0 (API 36.0)

Device : Pixel 9a

Assignment

Communication is an important matter. In our connected world, we have no choice but to be able to communicate. In order to help you understand how this works, you will build a chat application.

The server is provided and its documentation is provided.

Server address: mychat.fgontier.fr

The following features are **mandatory**:

- Authentication (login & register)
- List, join and leave a channel
- List channel's messages
- Send message to requested channel

Bonuses

The API you will be using provides more features available than the mandatory features. Use them as much as you want.

I intentionally included a security breach. The first group to find it will be given the maximum grade. In order to valid your discovery, you must send me a **private** message on Teams describing the security breach and how to exploit it. Once the breach has been validated, no other group can receive the maximum grade.

Timeline

Kick-off: 20/03/2026 - 13h00

Follow-up: 24/03/2026 - 9h30

Delivery: 16/04/2026 - 10h00

MyChat

General resources

In order to guide you, here are some knowledge you should learn in order to be able to complete the project.

REST communication

The server provided is a REST application. REST is the most common protocol when it comes to exchange information between multiple application.

Here are some resources about REST and HTTP requests in Android & Kotlin:

[What Is REST API?](#) (Youtube)

[Using Fuel to make HTTP request](#) (Tutorial)

Material UI

Material is a very popular UI library created by Google. Using Material will make it simple to create beautiful UI in a short amount of time.

[Guide to Material design for Android](#) (Tutorial)

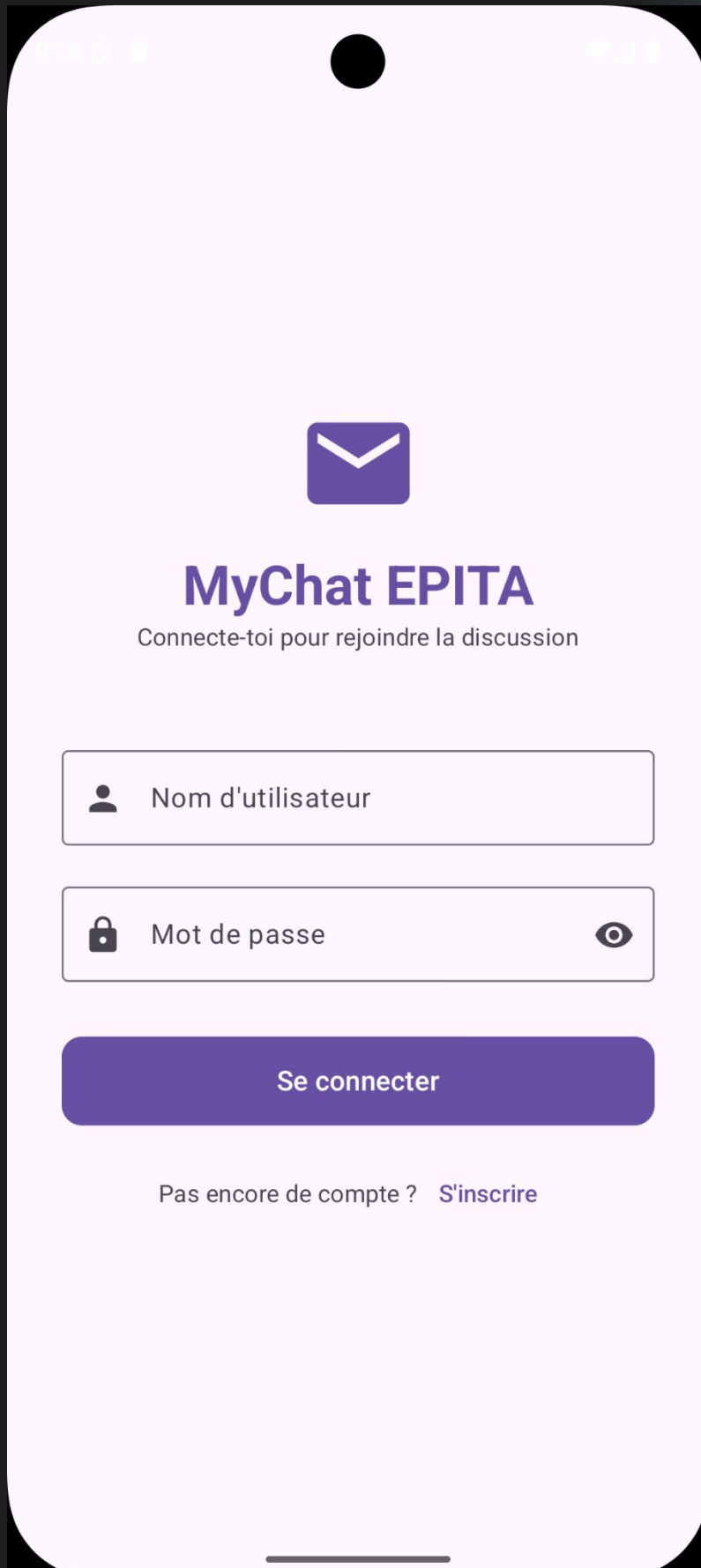
Step by step recommendation

1. Learn to send a HTTP request using Fuel
2. Create a register view which allows you to create an account to the server
3. Create a login view which allows you to login to the server

The server will give you a JWT token. Please, read about this.

4. Create a menu listing all channels available
5. Create a menu listing all messages from a channel
6. Add a textbox to be able to send a message to a given channel

UI example – login



A mobile app login screen for 'MyChat EPITA'. The screen is white with rounded corners and a black notch at the top. At the top, there is a status bar with the time '12:00', a signal strength indicator, and a battery level indicator. Below the notch is a purple envelope icon. The title 'MyChat EPITA' is in a bold, purple font, followed by the subtitle 'Connecte-toi pour rejoindre la discussion' in a smaller, gray font. There are two input fields: the first is for the username, labeled 'Nom d'utilisateur' with a user icon on the left; the second is for the password, labeled 'Mot de passe' with a lock icon on the left and an eye icon on the right to toggle visibility. Below the input fields is a large, rounded purple button with the text 'Se connecter' in white. At the bottom, there is a link that says 'Pas encore de compte ? S'inscrire'.

12:00



MyChat EPITA

Connecte-toi pour rejoindre la discussion

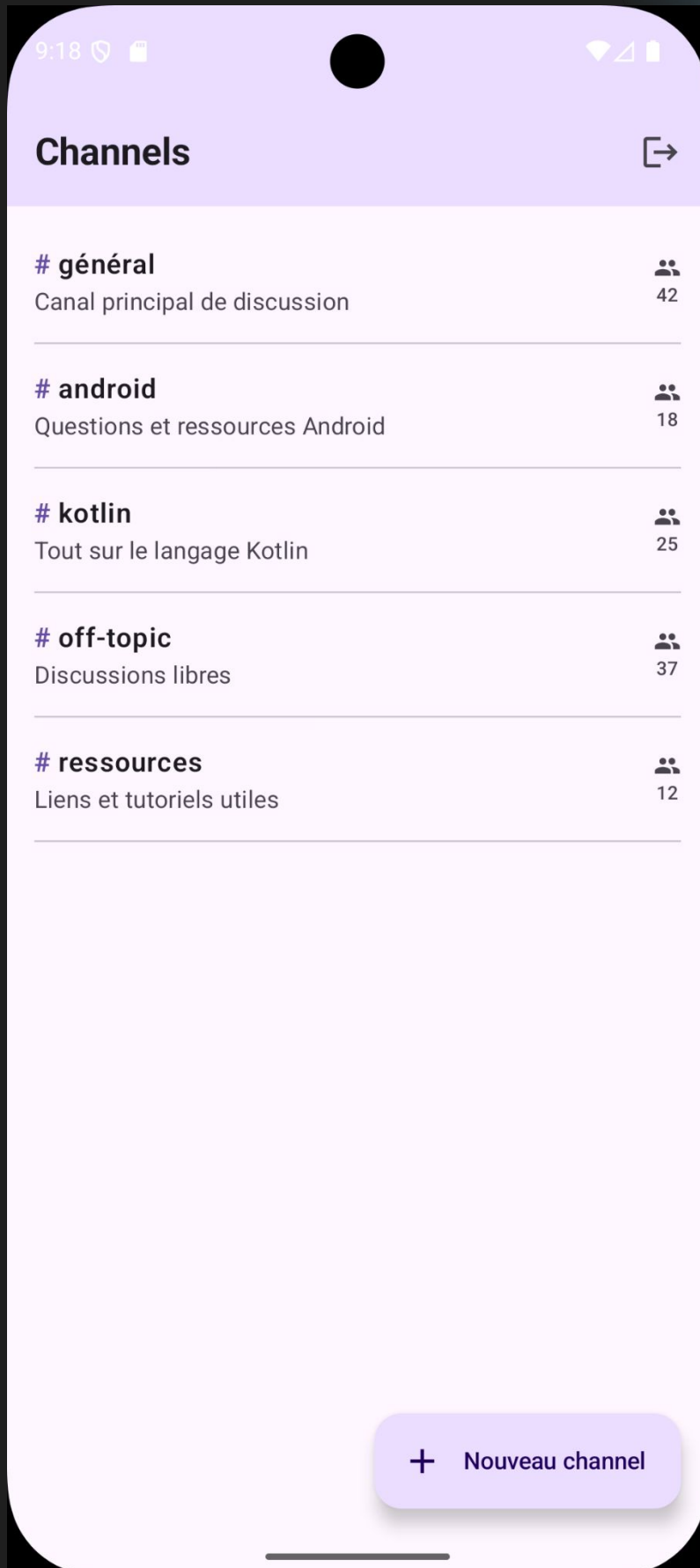
 Nom d'utilisateur

 Mot de passe 

Se connecter

Pas encore de compte ? [S'inscrire](#)

UI example – login



UI example – channel

